Ensemble Methods

$$\bar{h}(x) = \mathbb{E}_{P_D}\left[h(x)\right]$$

In Ensemble methods, $\mathbb{E}_{P_D}$ is approximated via creating datasets by sampling with replacement.

Bagging : Sample with replacement to create an Ensemble.

suppose the dataset D contain N samples
create B training sets by sampling
so that each new dataset D_b has N samples

prob. of a sample being picked $1/N$

prob. of $x_i \notin D_b = \left(1 - 1/N\right)^N$

$$\lim_{N \to \infty} \left(1 - 1/N\right)^N = \frac{1}{e} = 0.36$$

we train B i/d classifiers
$h_1(x), h_2(x) \ldots h_B(x)$

The final prediction:

$$\hat{y}_{Bag} = \frac{1}{B} \sum_{b=1}^{B} h_b(x).$$

The variance

suppose Z_i represent a Rv, corresponding to the prediction of the i^{th} hypothesis on a given datapoint.

Since all the hypo. are trained on the data sampled from a given distribution we assume $Z_1 \ldots Z_B$ to be identically distributed.

$\therefore \ var(Z_i) = \sigma^2$ for all i.

Since Z_i & Z_j are not stat. independent

$$corr(Z_i, Z_j) = \rho \quad \forall\ i \neq j$$

$$corr(\ ) = \frac{\text{covariance}}{\sigma_i \sigma_j}$$

$$\Rightarrow \quad \mathrm{cov}(z_i, z_j) = \rho \cdot \sigma^2$$

To calculate the var. of Ensemble

$$\bar{z} = \frac{1}{B} \sum_{i=1}^{B} z_i$$

$$\mathrm{var}(\bar{z}) = \mathrm{var}\left(\frac{1}{B} \sum_{i=1}^{B} z_i\right)$$

$$= \frac{1}{B^2} \mathrm{var}\left(\sum_{i=1}^{B} z_i\right)$$

$$\mathrm{var}\left(\sum_{i=1}^{B} z_i\right) = \sum_{i=1}^{B} \mathrm{var}(z_i) + \sum_{i \neq j} \mathrm{cov}(z_i, z_j)$$

$$\Rightarrow$$

$$\mathrm{var}(\bar{z}) = \rho\sigma^2 + \left(\frac{1-\rho}{B}\right)\sigma^2$$

To reduce $\mathrm{var}(\bar{z})$

i) Increase B

ii) reduce ρ : ensure that individual hypo. are uncorrelated

Eg: Random Forests:

Ensemble of Decision trees

a randomly sampled subset of data dimensions are choosen at each node of every individual tree to reduce "ρ".

Boosting algorithms.

The Ensemble is built, iteratively in an incremental manner.

Let $h_m(x)$ denote the Classifier learned at the m^{th} iteration.

for $m=1$ to B

$$H_m(x) = H_{m-1}(x) + \alpha_m h_m(x).$$

where $H_m(x)$ is the incremental Ensemble obtained at the m^{th} iteration.

<u>Question</u>: Given $H_{m-1}(x)$, how to train $h_m(x)$ so that loss is reduced?

Gradient Boosting mechanism

Recall,

$$H_m(x) = H_{m-1}(x) + \alpha_m h_m(x)$$

<u>Goal:</u> Find a new learner $h_m(x)$ to minimize the risk.

$$\underset{h,\alpha}{\operatorname{argmin}} \sum_{i=1}^{N} L\left(y_i, H_m(x)\right)$$

Consider

$$L\left(y_i, H_m(x_i)\right) = L\left(y_i, H_{m-1}(x_i) + \alpha_m h_m\right)$$

$$L\left(y_i, H_{m-1}(x_i) + \alpha_m h_m\right) \approx$$

$$L\left(y_i, H_{m-1}(x_i)\right) + \underbrace{\left[\frac{\partial L\left(y_i, H(x_i)\right)}{\partial H(x_i)}\right]_{H=H_{m-1}}}_{g_i} \cdot \alpha h(x_i)$$

↑ Taylor's series around H_{m-1}.

$$LHS = L\left(y_i, H_{m-1}\left(x_i\right)\right) + \alpha g_i h(x_i)$$

$$\underset{\alpha, h}{\operatorname{argmin}} \sum_{i=1}^{N} \left[\underbrace{L\left(y_i, H_{m-1}(x_i)\right)}_{\text{Ind of } h \,\&\, \alpha} + \alpha g_i h(x_i) \right]$$

$$\boxed{= \underset{\alpha, h}{\operatorname{argmin}} \sum_{i=1}^{N} \alpha \cdot g_i h(x_i)}$$

$g_i \big|_{H=H_{m-1}}$ can be computed.

question: How to minimize $\sum g_i h(x_i)$

note that $\sum g_i h(x_i) = \langle g, h \rangle$

$$\Rightarrow \underset{h}{\operatorname{argmin}} \langle g, h \rangle = h^* \,\big|\, h^* \propto -g_i$$

To find h^* that is proportional to $-g_i$: solve a regression problem with $-g_i$ as labels.

$$h_m = \operatorname{argmin} \sum_{i=1}^{N} \left[-g_i - h(x_i)\right]^2$$

The Gradient Boosting Algorithm.

Given a dataset D,

Start by creating B datasets (sampling with replacement)

for $m=1$ to $B-1$ do:

$$H_m(x) = H_{m-1}(x) + \alpha h_m(x)$$

compute $\underbrace{-g_i^m}_{\text{residual}} = -\left[\frac{\partial L(y_i, H(x_i))}{\partial H(x_i)}\right]_{H=H_{m-1}}$

Train $h_m(x)$ by regressing over g^m.

get $H_m(x) = H_{m-1}(x) + \alpha h_m(x)$

End For.